# Isaac ROS 3D Pose estimation

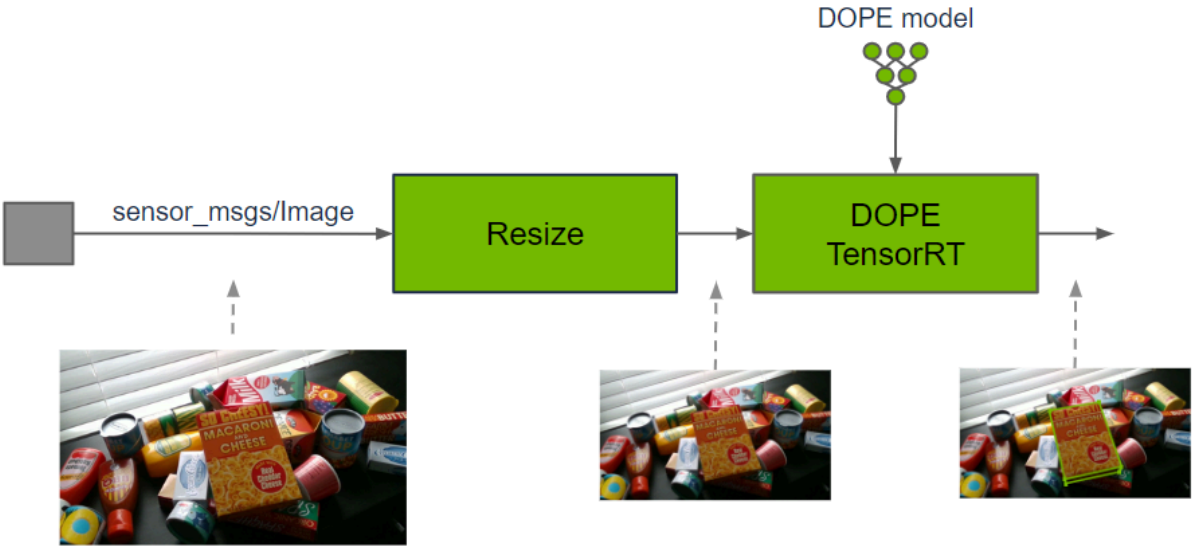
**Isaac ROS 3D Pose estimation**
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Isaac ROS CenterPose official website link: https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_pose_estimation/isaac_ros_centerpose/index.html

## Overview

Isaac ROS provides several object pose estimation solutions. CenterPose uses GPU-accelerated inference to predict the 3D position and pose of objects of known classes, and the result can be displayed together with the image and the camera information.

| Package | Characteristics |
|---|---|
| `isaac_ros_foundationpose` | Supports new objects without retraining |
| `isaac_ros_dope` | Pose estimation for known objects |
| `isaac_ros_centerpose` | Pose estimation that supports the TAO workflow |




## Precision choice of the NX model

On the Jetson Orin NX and in the current TensorRT environment, this case uses the FP32 engine: `/workspaces/isaac_ros-dev/isaac_ros_assets/centerpose_shoe_fp32.plan`. The script `scripts/centerpose_launch.sh` that comes with this tutorial already specifies `centerpose_shoe.onnx` and that FP32 plan, and sets `force_engine_update:=False`; a normal run does not require moving files between

the model directories, and you must not switch to FP16.

Before starting, you can confirm the model file and the choice made by the script:

```
ls -lh /workspaces/isaac_ros-dev/isaac_ros_assets/centerpose_shoe.onnx
/workspaces/isaac_ros-dev/isaac_ros_assets/centerpose_shoe_fp32.plan
rg -n 'model_file_path|engine_file_path|force_engine_update' /workspaces/isaac_ros-
dev/scripts/centerpose_launch.sh
```

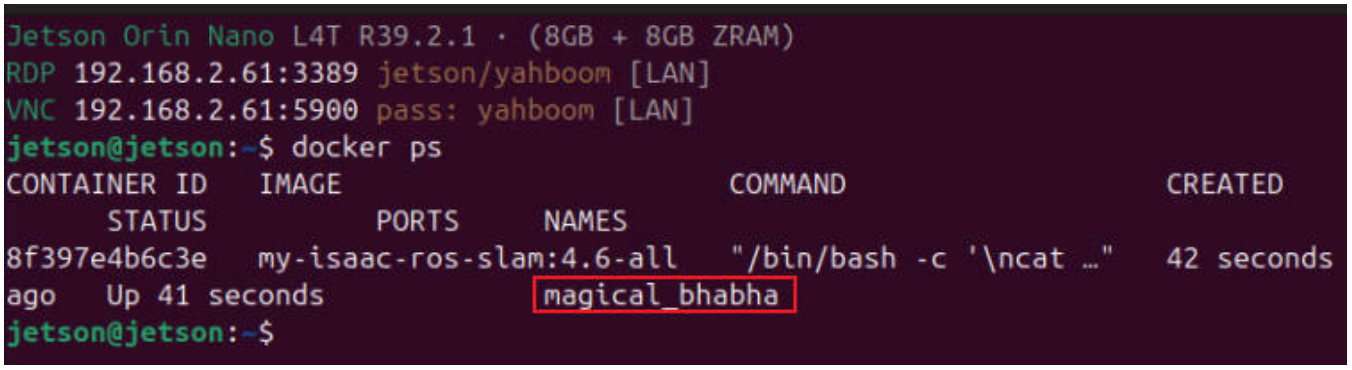
If the FP32 plan is missing, do not substitute FP16 or force a rebuild of the engine; contact the course provider for the model files that match this environment, and then run the case.

# Quick experience

First start the container with the following command

```
./run_isaac.sh
```

Then confirm the container on the host with `docker ps`, and open the container terminals with `docker exec -it <container name> bash`. Each terminal has to use the same ROS/FastDDS environment, for example:




**Note: all the terminals below are executed inside the container:**

Terminal A: start the inference:

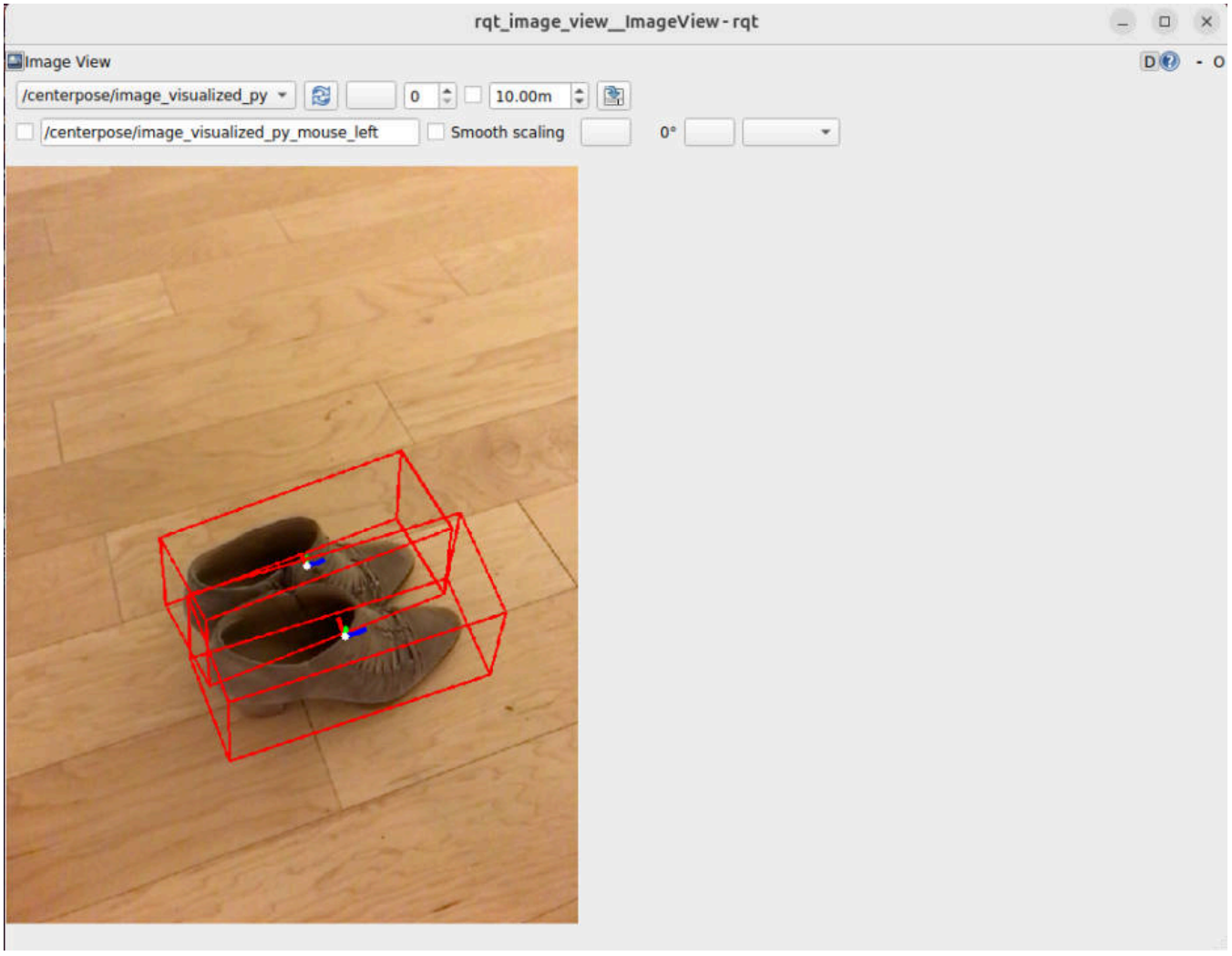
```
bash /workspaces/isaac_ros-dev/scripts/centerpose_launch.sh
```

Terminal B: start the two bag streams needed for the inference and the display:

```
bash /workspaces/isaac_ros-dev/scripts/centerpose_playback.sh
```

Terminal C: start the Python pose box visualisation and rqt:

```
bash /workspaces/isaac_ros-dev/scripts/centerpose_view.sh
```




Select the topic `/centerpose/image_visualized_py` in rqt. To check the output, run:

```
ros2 topic info /centerpose/detections -v
ros2 topic echo /centerpose/detections --once
ros2 topic hz /centerpose/image_visualized_py
```